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SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: MLA03
COURSE NAME: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO2: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr G.A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A
Application - Based Questions
Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality. (5 Marks)

Understanding of Problem Context
A ride-sharing company needs to match available drivers with passengers quickly. The main objectives are to **reduce passenger waiting time, reduce driver idle time, and improve service quality**.
Application of RL Concepts
The system can model the problem as a Reinforcement Learning environment:
- **State Space:** Current locations of drivers and passengers, driver availability, passenger waiting time, traffic conditions, and distance between driver and passenger.
- **Action Space:** Assign a particular driver to a passenger, keep the driver available, or assign another nearby driver.
- **Reward Function:**
  - Positive reward for successful and quick matching.
  - Negative reward for long passenger waiting time, long travel distance, or driver idle time.
- **Agent:** RL matching algorithm.
- **Environment:** Ride-sharing network containing drivers, passengers, roads, and traffic.

Problem-Solving Approach
The agent observes the current state, selects the best driver-passenger matching action, receives a reward, and updates its policy. Algorithms such as **Q-Learning or Deep Q-Network (DQN)** can be used for large state spaces.
Accuracy of Solution

A suitable reward function can be:

Reward = +10 for successful fast matching − waiting time penalty − unnecessary travel penalty.

The agent therefore learns to select drivers that can reach passengers quickly.

Clarity / Conclusion

Over repeated interactions, the RL agent learns an optimal **matching policy** that minimizes waiting time, reduces driver idle periods, and improves overall ride-sharing efficiency.

2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations. (5 Marks)

Understanding of Problem Context

An e-learning platform must recommend study materials according to each student's learning level and interests. The challenge is to balance **exploitation of materials already known to be useful** with **exploration of new materials**.

Application of Concepts

- **State:** Student's learning level, previous materials studied, quiz scores, interests, and learning progress.
- **Actions:** Recommend different videos, notes, quizzes, exercises, or new topics.
- **Reward:** Positive reward when the student completes content, obtains a higher quiz score, or shows improved engagement.

Problem-Solving Approach

The system uses two strategies:

Exploitation: Recommend materials that have already produced good results for the student.

Exploration: Try new materials that the student has not previously used.

For example, an **ε-greedy strategy** can select a known best recommendation most of the time while occasionally trying a new recommendation.

Accuracy of Solution

If $\varepsilon = 0.2$, the system can approximately:

- **80% of the time:** exploit the currently best-known recommendation.
- **20% of the time:** explore a new recommendation.

The reward from each recommendation updates the model.

Clarity / Conclusion

Thus, exploration prevents the system from repeatedly recommending the same content, while exploitation uses successful past recommendations. This produces **personalized and continuously improving learning recommendations**.

3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time. **(5 Marks)**

Understanding of Problem Context

A robotic vacuum cleaner must clean maximum floor area while avoiding obstacles such as walls, furniture, stairs, and objects.

Application of RL Concepts

- **State Space:** Robot's current position, battery level, obstacle locations, cleaned/un-cleaned areas, and sensor readings.
- **Action Space:** Move forward, backward, left, right, turn, stop, or return to charging station.
- **Reward:**
    - Positive reward for cleaning a new area.
    - Negative reward for hitting obstacles.
    - Negative reward for unnecessary movement.

- Positive reward for returning safely to the charging station.

Policy

A **policy π(s)** defines the action the robot should take for a particular state.
For example:
If an obstacle is detected in front → turn left or right.
If battery is low → return to charging station.
The policy becomes better as the robot learns from previous cleaning experiences.

Value Function

The value function V(s) estimates the expected future reward from a particular state.
A state where the robot is close to an uncleaned area and has sufficient battery may have a high value, while a state near an obstacle with low battery may have a lower value.

Conclusion

By updating its value function and policy after every cleaning episode, the robot gradually learns shorter, safer, and more efficient cleaning routes.

4. **A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance. (5 Marks)**

Understanding of Problem Context

A cryptocurrency trading system must decide whether to **Buy, Sell, or Hold** based on changing market conditions. Cryptocurrency prices are highly volatile, making reward design difficult.

Application of RL Concepts

- **State:** Current cryptocurrency price, price change, trading volume, technical indicators, portfolio balance, and previous action.
- **Actions:** Buy, Sell, or Hold.
- **Reward:** Change in portfolio value after considering transaction costs and risk.

A simplified reward can be:

Reward = Profit/Loss − Transaction Cost − Risk Penalty

Problem-Solving Approach

The RL agent observes the market state, chooses an action, receives the resulting profit or loss, and updates its policy.

Reward Design Challenges

Reward design is difficult because:

1. Short-term profit may encourage excessive trading.
2. Ignoring risk may cause the agent to take dangerous positions.
3. Transaction costs can reduce actual profit.
4. A reward based only on immediate profit may ignore long-term performance.
5. Highly volatile markets can produce unstable learning.

Accuracy / Conclusion

Poor reward design can make the agent maximize profit in the short term while taking excessive risk. Therefore, the reward should consider profit, transaction cost, risk, and long-term portfolio performance.

5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism. (5 Marks)

Understanding of Problem Context

A smart grid must distribute electricity efficiently while balancing electricity generation, consumption, storage, and demand. RL can dynamically learn how to allocate available electricity.

Application of RL Concepts

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State Space:

- Current electricity demand
- Power generation
- Battery/storage level
- Electricity price
- Renewable energy availability
- Grid load

Action Space:

- Increase/decrease power supplied to different areas
- Charge/discharge batteries
- Use renewable or conventional energy
- Adjust energy distribution

Reward Function:

- Positive reward for satisfying demand efficiently.
- Negative reward for power wastage.
- Negative reward for overload.
- Negative reward for high operating cost.

Problem-Solving Approach

At every time step, the RL agent observes the grid state and chooses an energy-distribution action. The resulting power balance and cost are used to calculate the reward.

Accuracy / Conclusion

The objective can be represented as:

Reward = Demand Satisfaction − Energy Cost − Power Loss − Overload Penalty

Through repeated learning, the system develops a policy that provides reliable electricity distribution with lower cost and energy wastage.

6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes. (5 Marks)

Understanding of Problem Context

A healthcare monitoring system can use RL as a **clinical decision-support system** to help determine how treatment should be adjusted according to a patient's changing condition. RL has been studied for sequential treatment and dosage optimization, but clinical deployment requires strong safety controls and clinician oversight.

Application of RL Concepts

State Space:

- Patient vital signs
- Relevant laboratory measurements
- Current symptoms
- Previous medication and response
- Patient history and other clinically relevant information

Action Space:

- Maintain current treatment
- Adjust treatment within clinically approved ranges
- Change treatment according to predefined medical protocols

Reward:

- Positive reward for improvement in relevant health outcomes.
- Negative reward for adverse effects.
- Negative reward for unsafe treatment changes.

Problem-Solving Approach

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The agent observes the patient's current state, selects a safe treatment recommendation, observes the patient's response, and updates its learned policy. In practical healthcare systems, RL should be constrained by clinical rules and evaluated carefully before real-world use.

Accuracy of Solution

The reward must balance treatment effectiveness and safety rather than simply maximizing one measurement. Incorrect reward design could encourage unsafe actions.

Conclusion

RL can help personalize sequential treatment decisions by learning from patient responses, but it should function as decision support under clinical supervision, not as an unrestricted autonomous prescriber.

7. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization. (5 Marks)

Understanding of Problem Context

A delivery drone needs to travel from a warehouse to a customer while minimizing travel time and energy consumption and avoiding obstacles and restricted areas.

Application of Concepts

- **State:** Drone position, altitude, battery level, obstacle locations, weather conditions, destination, and current route.
- **Actions:** Move forward, backward, left, right, up, down, or change direction.
- **Reward:**
  - Positive reward for reaching the destination.
  - Negative reward for long routes.
  - Negative reward for high energy consumption.
  - Large negative reward for collisions or entering restricted areas.

Exploration and Exploitation

Exploration means trying different routes to discover potentially better paths.

Exploitation means selecting the best route already learned from previous experience.

For example, using an ε-greedy strategy, the drone mostly selects the best-known route but occasionally explores another safe route.

Problem-Solving Approach

The drone learns route quality from repeated delivery trips. If a new route produces less travel time and energy consumption, its value increases.

Conclusion

Balanced exploration and exploitation allow the drone to discover shorter and safer routes while eventually exploiting the most efficient route.

8. Design and develop a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency. (5 Marks)

Understanding of Problem Context

A manufacturing robot performs repeated assembly operations. The goal is to complete each product quickly, accurately, and safely while minimizing unnecessary movements.

Application of RL Concepts

- **State:** Robot position, component position, assembly stage, tool status, and sensor readings.
- **Actions:** Pick component, move arm, rotate component, place component, tighten/assemble, or move to the next stage.
- **Reward:**
  - Positive reward for correct assembly.
  - Negative reward for incorrect placement.
  - Negative reward for excessive movement.

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- Negative reward for damaged components or unsafe operations.

Policy
The policy determines the best action for each assembly state.
For example:
If component A is correctly positioned → pick component B.
If assembly is complete → move to the next workstation.
Value Function
The value function estimates the future reward of each state. States that lead to fast and correct assembly receive higher values.
The robot updates its value estimates after every operation and gradually learns the most efficient sequence.
Conclusion
The combination of policy and value function enables the manufacturing robot to improve its assembly sequence, reduce errors, minimize movement, and increase production efficiency.

9. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance.
(5 Marks)

Understanding of Problem Context
An online advertising system must decide which advertisement to display to a user in order to maximize **click-through rate (CTR)** while maintaining user engagement and avoiding excessive or irrelevant advertisements.
Application of RL Concepts

- **State:** User interests, browsing behavior, previous ad interactions, time, device, and current webpage.
- **Actions:** Display different advertisements or choose not to display an advertisement.
- **Reward:** Positive reward for useful clicks and engagement; negative reward for irrelevant ads or poor user experience.

Problem-Solving Approach
The agent displays an advertisement, observes whether the user clicks or interacts with it, and receives a reward. It then updates its policy to improve future recommendations.
Reward Design Challenges
A reward based only on clicks can cause problems:

1. The system may show misleading or sensational advertisements.
2. It may maximize clicks but reduce user satisfaction.
3. Repeated advertisements may cause user fatigue.
4. Short-term CTR may increase while long-term engagement decreases.

Accuracy / Conclusion
A better reward should consider:
Reward = CTR + User Engagement − Ad Fatigue − Negative Feedback
Therefore, proper reward design is essential because the RL agent optimizes exactly what the reward function measures.

10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function. **(5 Marks)**

Understanding of Problem Context
A smart irrigation system must provide enough water for healthy crop growth while avoiding water wastage. RL can learn irrigation decisions from soil and environmental conditions.
Application of RL Concepts
State Space
The state can contain:

- Soil moisture level
- Temperature
- Humidity
- Rainfall/weather forecast
- Crop growth stage
- Water tank level
- Previous irrigation activity

Action Space

The agent can:

- Turn irrigation OFF
- Turn irrigation ON
- Select low, medium, or high water quantity
- Change irrigation duration

Reward Function

The reward should encourage healthy crops and efficient water use:

Reward = Crop Health Benefit − Water Consumption − Over-Irrigation Penalty

For example:

- Healthy soil moisture → positive reward.
- Crop receives required water → positive reward.
- Excessive water usage → negative reward.
- Very dry soil → large negative reward.

Problem-Solving Approach

The agent observes the current soil and weather conditions, chooses an irrigation action, observes the resulting soil moisture and crop condition, and updates its policy.

Accuracy / Conclusion

After repeated learning, the system can identify when and how much water should be supplied. Thus, RL helps achieve efficient irrigation, reduced water wastage, and better crop management.

Code of Conduct:

I P. Sahith Sai (192425029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Sahith Sai

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand